

Problem A. Fast Power

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

Given two integers x and k ($1 \leq x, k \leq 10^9$), please compute x^k modulo 998244353.

Input

Two integers x and k in one line.

Output

Please output the answer.

Examples

standard input	standard output
5 8	390625
13 2153	687278077

Problem B. Inversion

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

Given a permutation a of length n , please compute the number of pairs (i, j) such that $i < j$ and $a_i > a_j$. Note that a permutation means that each element from 1 to n occur exactly once.

Input

The first line contains an integer T ($1 \leq T \leq 5$), indicating the number of testcases.

For each testcase, the first line contains an integer n ($1 \leq n \leq 5 \times 10^5$), the second line contains a permutation of 1 to n .

Output

Output the answer.

Example

standard input	standard output
2	3
3	7
3 2 1	
5	
5 2 3 4 1	

Problem C. Maximum Subarray

Input file: `standard input`
Output file: `standard output`
Time limit: 1 second
Memory limit: 256 megabytes

Given a sequence of length n , please compute the subarray of the largest sum. You only need to output the sum. Note that the subarray can be empty.

Input

There is only one test case in each test file.

The first line of the input contains an integer n ($1 \leq n \leq 10^6$) indicating the length of the sequence.

The second line of the input contains n integers $a_1, a_2, \dots, a_n$ ($-10^6 \leq a_i \leq 10^6$) indicating the original sequence.

Output

Output the answer in one line.

Example

standard input	standard output
5 -5 7 9 -1 3	18

Problem D. Binary Tree

Input file: `standard input`
Output file: `standard output`
Time limit: `1 second`
Memory limit: `256 megabytes`

Teacher JB first wrote down two strings for each binary tree: a preorder traversal (root, left subtree, right subtree) and an inorder traversal (left subtree, root, right subtree).

We all know that such a pair of strings would give enough information to reconstruct the tree later.

Now it is your job to write a program that does the job for him!

Input

The input will contain one or more test cases.

Each test case consists of one line containing two strings `preord` and `inord`, representing the preorder traversal and inorder traversal of a binary tree. Both strings consist of unique capital letters. (Thus they are not longer than 26 characters.)

Input is terminated by end of file.

Output

For each test case, recover the binary tree and print one line containing the tree's postorder traversal (left subtree, right subtree, root).

Example

<code>standard input</code>	<code>standard output</code>
<code>ABC BAC</code>	<code>BCA</code>

Problem E. Nearest Pairs

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

Given n points $P(x, y)$, please compute

$$\min_{i \neq j} |P_i - P_j|$$

Input

The first line contains an integer n ($n \leq 10^5$).

The following n lines contain two integers x and y for each point.

Output

Please output the answer. Your answer will be considered correct if and only if the error is within 1E-6.

Examples

standard input	standard output
6 24 58 39 67 18 57 92 83 75 10 5 7	6.08276253029821933893
5 4 8 5 2 3 7 6 10 1 3	1.41421356237309514547